# Homework 1

2026-09-10

## The Data

```
study_time <- c(12, 15, 18, 20, 22, 22, 25, 27, 28, 30,
                30, 31, 33, 35, 36, 38, 40, 42, 45, 50)
study_time
```

```
##  [1] 12 15 18 20 22 22 25 27 28 30 30 31 33 35 36 38 40 42 45 50
```

---

## 8.

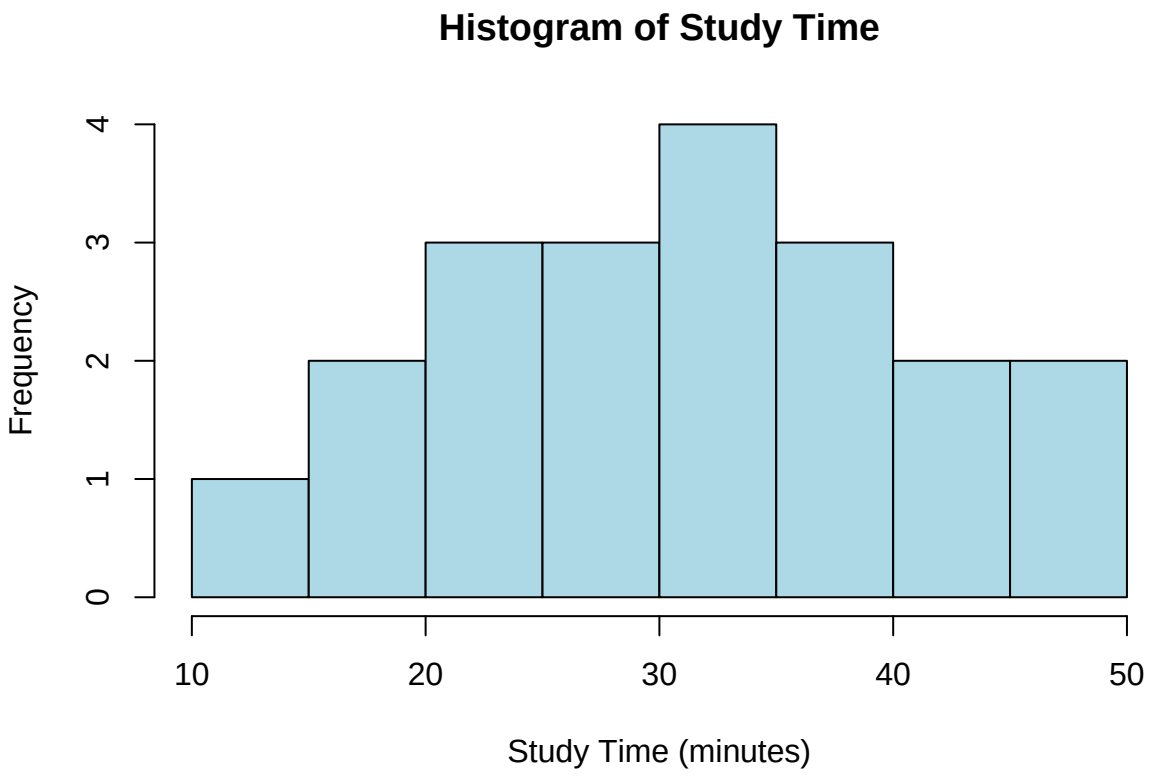
### (a) Frequency Table

```
breaks <- seq(10, 54, by = 5)
bins <- cut(study_time, breaks = breaks, right = FALSE)
freq_table <- table(bins)
freq_table
```

```
## bins
## [10,15) [15,20) [20,25) [25,30) [30,35) [35,40) [40,45) [45,50)
##       1       2       3       3       4       3       2       1
```

### (b) Histogram

```
hist(study_time,
     breaks = breaks,
     right = FALSE,
     col = "lightblue",
     border = "black",
     main = "Histogram of Study Time",
     xlab = "Study Time (minutes)",
     ylab = "Frequency")
```




### (c) Shape of the Distribution

The distribution is close to symmetric, but it slightly skews right.

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## 9. Box Plot

### (a) Five Number Summary

```
summary(study_time)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   12.00   22.00   30.00   29.95   36.50   50.00
```

```
Q1 <- quantile(study_time, 0.25)
Q3 <- quantile(study_time, 0.75)
med <- median(study_time)
min_val <- min(study_time)
max_val <- max(study_time)

c(Min = min_val, Q1 = Q1, Median = med, Q3 = Q3, Max = max_val)
```

```
##    Min Q1.25% Median Q3.75%    Max
##   12.0   22.0   30.0   36.5   50.0
```

### (b) IQR and Outlier Check

```
IQR_val <- Q3 - Q1
lower_fence <- Q1 - 1.5 * IQR_val
upper_fence <- Q3 + 1.5 * IQR_val

IQR_val
```

```
##  75%
## 14.5
```

```
lower_fence
```

```
##  25%
## 0.25
```

```
upper_fence
```

```
##   75%
## 58.25
```

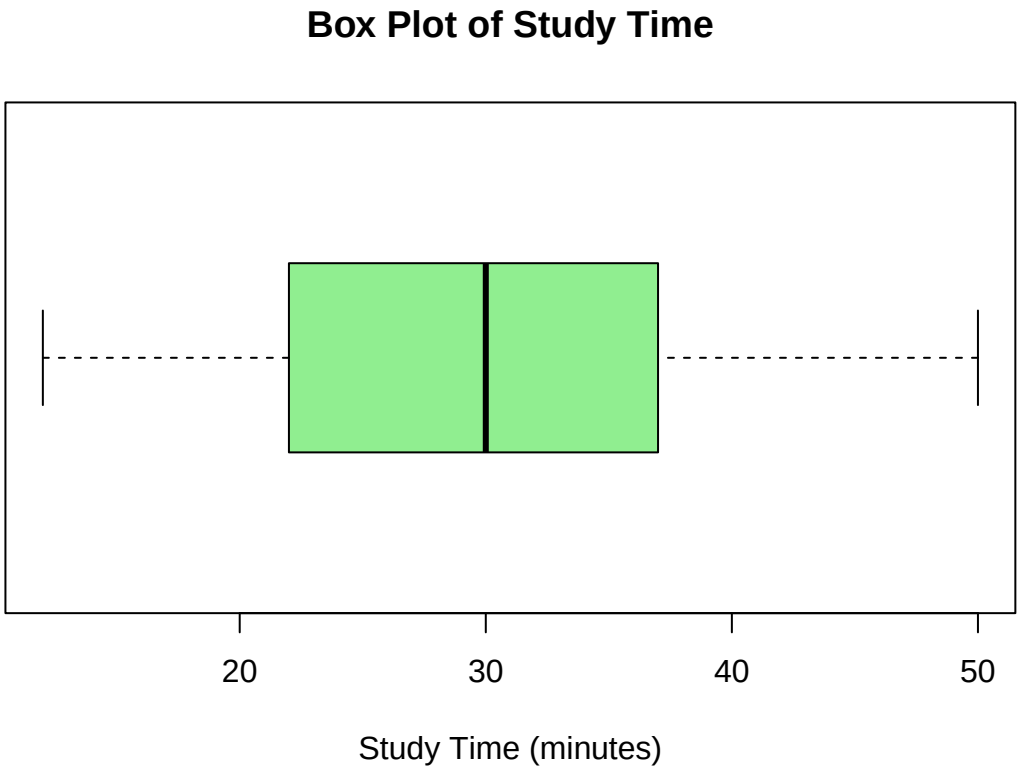
```
outliers <- study_time[study_time < lower_fence | study_time > upper_fence]
outliers
```

```
## numeric(0)
```

numeric(0) means no outliers

### (c) Box Plot

```
boxplot(study_time,
        horizontal = TRUE,
        col = "lightgreen",
        main = "Box Plot of Study Time",
        xlab = "Study Time (minutes)")
```




---

## 10. Comparison

### (a) Mean vs. Median

```
mean(study_time)
```

```
## [1] 29.95
```

```
median(study_time)
```

```
## [1] 30
```

I would say the median is the best measure of the center. The histogram showed a large amount of instances between 30 and 35. The median came out to exactly 30.